



JÖNKÖPING UNIVERSITY
School of Engineering

TARI29 Artificial Intelligence, 7.5 credits

Study Guide

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Canvas course room: TARI29 T4322

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1 Teachers

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2 Introduction

The course is a basic course in artificial intelligence with a focus on traditional AI, i.e., GOFAI. The course covers many different basic and intermediate topics in the field, alternating theory with practice. After completing the course, the student shall have acquired a basic, but broad, knowledge in the field of artificial intelligence. Specifically, the student should understand and know how to apply all the theoretical concepts covered.

3 Intended Learning Outcomes

After a successful course, the student shall

Knowledge and understanding

- Display knowledge of search algorithms for production systems
- Display knowledge of formalisms for reasoning under uncertainty
- Show familiarity with game theory and its applications in AI

Skills and abilities

- Demonstrate the ability to apply production systems, and the accompanying search algorithms to various representations and problem instances
- Demonstrate the ability to develop computer programs that handle the concepts addressed in the course, including evolutionary computation
- Demonstrate the ability to use resolution for inferencing in predicate and propositional logic
- Demonstrate the ability to implement the Minimax algorithm, including some standard improvements
- Demonstrate the ability to implement algorithms for finding optimal policies in Markov decision processes

Judgement and approach

- Demonstrate an understanding of how representations and algorithms can be used for solving basic AI-problems described in a formal or informal way

4 Examination, ILOs and Learning Activities

4.1 Realization of the Intended Learning Outcomes

This course emphasizes *independent work*, using the course books, as well as alternative sources. While the lectures cover the most important parts, it is imperative that the students work actively with the material, as presented in the course books.

The course uses the Python programming language. There will be a limited number of exercise sessions with tutoring. However, students are expected to spend enough time learning the basics of Python programming.

The teaching events consist of 17 lectures covering different topics and two practical exercise sessions that provide support to learn the basics of Python programming for several selected problems. A workshop in academic writing is provided as additional aid. The lectures, exercises and workshop will be taught on campus. Each lecture is 2×45 minutes. The PDF slides from the lecture will be available for download from Canvas. The lecture slides will be added continuously by the teachers as the course is running. The exercise sessions will be conducted groups-wise. Consult Canvas to learn which group you belong to and then check the schedule to know what time and room your group session will be conducted in. The exercises will include solving selected problems with the help of Python. Teacher's assistance will be provided during the exercises. The exercise material will be added continuously by the teachers as the course is running.

The table in Section 7 presents the different topics with relevant textbook chapters and indicates a suitable order to follow the lectures and exercises.

4.2 Assignment

The first part of the course examination is an assignment. The assignment work is carried out in groups of 4–5 students, and you perform group division yourself in Canvas. The assignment comprises 2.5 credits and has the grading scale 5, 4, 3 or Fail. The task and its organisation are described in a separate document. The work includes coding in a programming language and writing a report. The assignment will end with a short viva (oral examination).

The assignment falls within the area of evolutionary optimisation. For the assignment work, the students are free to use any imperative or object-oriented programming language. Prolog or functional programming languages (e.g., LISP or Haskell) may not be used for these tasks.

There are three examination occasions, where the first is in October 2026, the second in December 2026 and the last one is in August 2027. Students who have not obtained a grade for the assignment on any of these occasions will have to carry out the assignment work the next time the course is given.

4.3 Written Examination

The second part of the course examination is a written exam. The written exam comprises 5 credits and has the grading scale 5, 4, 3 or Fail.

The written exam will be performed using the Inspera digital examination platform Inspera. The written exam includes both theoretical questions and practical problems. Samples of exam questions will be provided in Canvas.

There are three examination occasions, where the first is in October 2026, the second in December 2026 and the last one is in August 2027.

5 Grading Criteria

The course is graded 5, 4, 3 or Fail. The final grade for the course is based on the written examination and assignment. The final grade will only be issued after satisfactory completion of *both* examination *and* assignment.

Examination The maximum total score for the written examination is 70 points, and the grading is:

3	28–41.5	points
4	42–55.5	points
5	56–70	points

Assignment The maximum score for the assignment is 30 points, and the grading is as follows:

3	12–17.5	points
4	18–23.5	points
5	24–30	points

Course grade The scores from the written examination and assignment are added to produce one final score, which determines course grade. The maximum total score is 100 points, and the grading is:

3	40–59.5	points
4	60–79.5	points
5	80–100	points

6 Course Literature and Other Teaching Aids

Textbooks The main textbook is:

Artificial Intelligence a modern approach,

Author: Russel, S. & Norvig, P.

4th Ed., ISBN: 978-1292401133

3rd Ed., ISBN: 978-0132071482

7 Time Plan

No	Topic	Reading	Teacher
Week 36			
A1	Course intro & Agents	Study guide, 1–2	VT
A2	Uninformed search	3.1–3.4	VT
Week 37			
A3	Informed search	3.5	VT
Ass.	Group assignment starts		VT
P1	Python crash course		JSS
E1	Python basics		JSS
Week 38			
A4	Non-traditional search	4.1	UJ
A5	Propositional logic: basics	7.1–7.4	VT
A6	Propositional logic: inference and agents	7.5–7.7	VT
P2	Python for AI		JSS
Ass.	Assignment hackathon		
Week 39			
A7	First-order logic: basics	8	VT
A8	First-order logic: inference	9	VT
Workshop	Academic Writing workshop		JS
Ass.	Group assignment tutoring		VT
Week 40			
A9	Reasoning under uncertainty I	12 (13)	VT
A10	Reasoning under uncertainty II	13, p.477 (14.7)	VT
A11	Language	TBD	JSS
E2	Python for AI		JSS
Week 41			
A12	Utility theory and decision making	15.1–15.3, 16.1–16.3 (16.1–16.2.2, 17.1–17.3)	VT
A13	Learning	23.1–23.4 (21.1–21.4)	JSS
Ass.	Group assignment deadline and viva		VT
Week 42			
A14	Adversarial search	6.1–6.3, 6.5–6.6 (5.1–5.6)	VT
A15	Game theory	17.2.3–17.2.4 (17.5) + notes	VT
Week 43			
19 October	Examination		

Teachers: VT (Vladimir Tarasov), JSS (Joel Scarinius Ståvmo) UJ (Ulf Johansson), SJ (Jenny Siméus). The chapters for the AI lectures are in Russell & Norvig's book. The chapters from the 3rd edition are given in parentheses when they differ from the 4th edition. The chapters for Prolog are in Bratko's book.